1. the main purpose of Quality of service in the networking is allows us to prioritize certain types of traffic over others, so that important receive the bandwidth, low delay, and reliability they need

2. jitter is the variation of delay of received packets. High jitter can cause voice packets to arrive out of order or too late, resulting in choppy audio, distortion, and poor call quality

3. queuing delay is the variable amount of time a frame or packet waits to be transmitted on the link, then serialization delay is the fixed amount of time it takes to transmit a frame onto the wire

4. I would check latency, jitter, packet loss, and bandwidth. Latency causes communication delays, jitter causes choppy video and audio, packet loss leads to poor image quality, and insufficient bandwidth can cause buffering and reduced video quality

5. The basic one-way QoS requirements for voice traffic are: latency less than 150 ms, jitter less than 30 ms, packet loss less than 1%, and sufficient bandwidth (typically 80–100 kbps per call including overhead).

6. Data traffic is relatively insensitive to drops and delays compared to voice and video. Voice traffic is predictable and smooth and very sensitive to delays and dropped packets. Video traffic tends to be unpredictable, inconsistent, and bursty. Compared to voice, video is less resilient to loss and has a higher volume of data per packet.

7.
=First In First Out (FIFO)
- Packets are buffered and forwarded in the order they arrive.
- FIFO has only one queue and treats all packets equally without prioritization.
=Priority Queuing (PQ)
- Traffic is placed into different priority queues.
- Higher-priority traffic is transmitted before lower-priority traffic.
=Weighted Fair Queuing (WFQ)
- Automatically allocates bandwidth fairly among different traffic flows.
- Applies weights to traffic so that more important traffic receives more bandwidth.
=Class-Based Weighted Fair Queuing (CBWFQ) / Low Latency Queuing (LLQ)
- CBWFQ: Allows user-defined traffic classes, with a separate FIFO queue for each class.
- LLQ: Extends CBWFQ by adding a strict priority queue for delay-sensitive traffic such as voice.

8. The lag may be caused by high latency, jitter, packet loss, or improper traffic handling, even if the network is lightly used. QoS can prioritize VoIP traffic, reduce delay and jitter, and ensure sufficient bandwidth for voice packets, improving call quality.

9. I would recommend traffic classification and marking, CBWFQ, and LLQ. Traffic classification and marking identify voice, video, and data traffic. CBWFQ allocates bandwidth to different traffic classes, while LLQ gives strict priority to delay-sensitive VoIP traffic. These QoS tools help reduce latency, jitter, and packet loss, improving call quality during peak hours.

10. The three main QoS models are Best-Effort, Integrated Services (IntServ), and Differentiated Services (DiffServ).

*Best-Effort treats all packets equally and provides no QoS guarantees.
*IntServ provides end-to-end QoS by reserving resources for individual flows. It uses admission control and a connection-oriented approach to guarantee bandwidth and delay requirements.
*DiffServ classifies traffic into different classes and applies QoS policies on a hop-by-hop basis. It is more scalable than IntServ but does not provide strict end-to-end guarantees.

11. DiffServ classifies traffic into different classes and assigns different priorities. Large email attachments can be placed in a lower-priority class, while critical applications receive higher priority. This improves performance and reduces congestion without reserving bandwidth for each user.